

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Digital Signal and Image Processing (01CT1513)	Aim : Compute and analyze the correlation between three audio tracks	
Open Ended Problem : 3	Date: 01/08/2026	Enrolment No: 92400133125

Aim : Compute and analyze the correlation between three audio tracks

- **Cross-Correlation**

Cross-correlation is used to compare two different signals. It helps determine whether the signals contain similar information and whether one signal is shifted with respect to the other. A higher peak in the cross-correlation graph indicates greater similarity between the two signals.

- **Auto-Correlation**

Auto-correlation compares a signal with itself. It is mainly used to identify repeating patterns or periodicity within the signal. The highest value of auto-correlation always occurs at zero lag, because every signal is identical to itself.

An audio signal is represented digitally as a sequence of amplitude samples. When two songs contain similar musical structures, their waveforms also become similar, resulting in a higher correlation.

In this experiment:

- **Original Song:** Contains vocals and instrumental music.
- **Karaoke Version:** Contains only instrumental music.
- **Different Song:** Contains different melody, rhythm, tempo, and instruments.

Because the original and karaoke versions share the same background music, they are expected to produce a higher correlation. On the other hand, the different song should produce a much lower correlation because its waveform pattern is unrelated.

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Digital Signal and Image Processing (01CT1513)	Aim : Compute and analyze the correlation between three audio tracks	
Open Ended Problem : 3	Date: 01/08/2026	Enrolment No: 92400133125

Code 1 :

```

import os

import numpy as np

import librosa

import matplotlib
matplotlib.use("TkAgg")

import matplotlib.pyplot as plt from scipy.signal

import correlate

base_dir = os.path.dirname(os.path.abspath(file))

original_file = os.path.join(base_dir, "Balam_Pichkari_Original.mp3")

karaoke_file = os.path.join(base_dir, "Balam_Pichkari_karaoke.mp3")

different_file = os.path.join(base_dir, "Aayi_Nai_-Stree_2.mp3")

for file in [original_file, karaoke_file, different_file]:

if not os.path.exists(file): print("File not found:", file)

raise FileNotFoundError(file)

sr = 22050

original, _ = librosa.load(original_file, sr=sr, mono=True)

karaoke, _ = librosa.load(karaoke_file, sr=sr, mono=True)

different, _ = librosa.load(different_file, sr=sr, mono=True)

length = min(len(original), len(karaoke), len(different))

```

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Digital Signal and Image Processing (01CT1513)	Aim : Compute and analyze the correlation between three audio tracks	
Open Ended Problem : 3	Date: 01/08/2026	Enrolment No: 92400133125

```
original = original[:length]
```

```
karaoke = karaoke[:length]
```

```
different = different[:length]
```

```
original = original / np.max(np.abs(original))
```

```
karaoke = karaoke / np.max(np.abs(karaoke))
```

```
different = different / np.max(np.abs(different))
```

```
def normalized_correlation(x, y): corr = correlate(x, y, mode="full") value =  
np.max(np.abs(corr)) / (np.linalg.norm(x) * np.linalg.norm(y)) return corr, value
```

```
corr1, val1 = normalized_correlation(original, karaoke) corr2, val2 =  
normalized_correlation(original, different) corr3, val3 = normalized_correlation(karaoke,  
different)
```

```
auto_original = correlate(original, original, mode="full") auto_karaoke = correlate(karaoke,  
karaoke, mode="full") auto_different = correlate(different, different, mode="full")
```

```
plt.figure(figsize=(10, 6))
```

```
plt.subplot(3, 1, 1)
```

```
plt.plot(auto_original)
```

```
plt.title("Autocorrelation - Original")
```

```
plt.subplot(3, 1, 2)
```

```
plt.plot(auto_karaoke)
```

```
plt.title("Autocorrelation - Karaoke")
```

```
plt.subplot(3, 1, 3)
```

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Digital Signal and Image Processing (01CT1513)	Aim : Compute and analyze the correlation between three audio tracks	
Open Ended Problem : 3	Date: 01/08/2026	Enrolment No: 92400133125

```
plt.plot(auto_different)
```

```
plt.title("Autocorrelation - Different")
```

```
plt.tight_layout()
```

```
plt.show()
```

```
plt.figure(figsize=(10, 6))
```

```
plt.subplot(3, 1, 1)
```

```
plt.plot(corr1)
```

```
plt.title("Original vs Karaoke")
```

```
plt.subplot(3, 1, 2)
```

```
plt.plot(corr2)
```

```
plt.title("Original vs Different")
```

```
plt.subplot(3, 1, 3)
```

```
plt.plot(corr3)
```

```
plt.title("Karaoke vs Different")
```

```
plt.tight_layout()
```

```
plt.show()
```

```
pairs = [ "Original\nKaraoke", "Original\nDifferent", "Karaoke\nDifferent" ]
```

```
values = [val1, val2, val3]
```

```
plt.figure(figsize=(6, 4))
```

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Digital Signal and Image Processing (01CT1513)	Aim : Compute and analyze the correlation between three audio tracks	
Open Ended Problem : 3	Date: 01/08/2026	Enrolment No: 92400133125

```
plt.bar(pairs, values)

plt.title("Normalized Correlation Comparison")

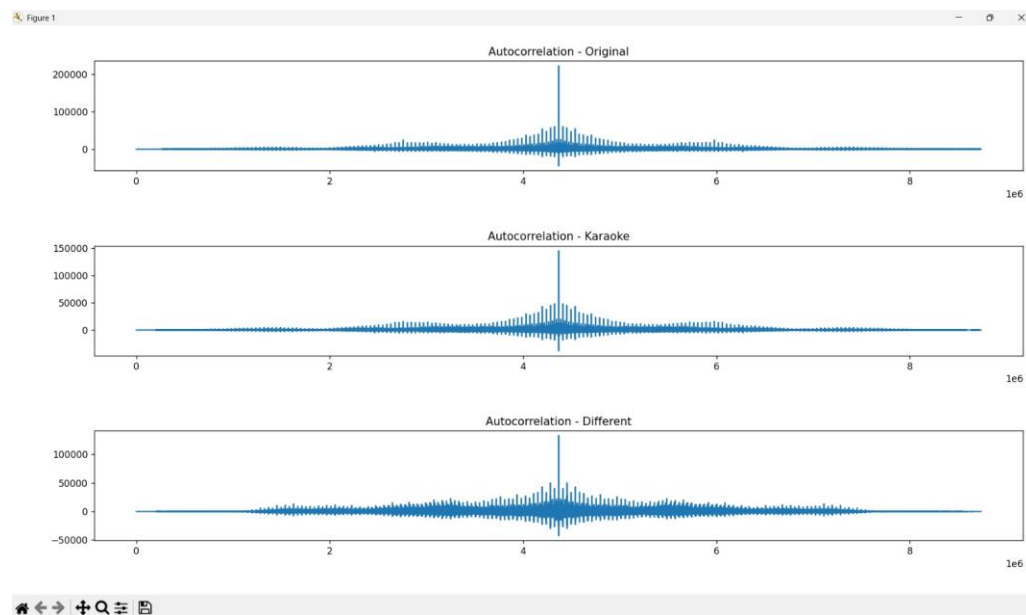
plt.ylabel("Correlation Value")

plt.ylim(0, 1)

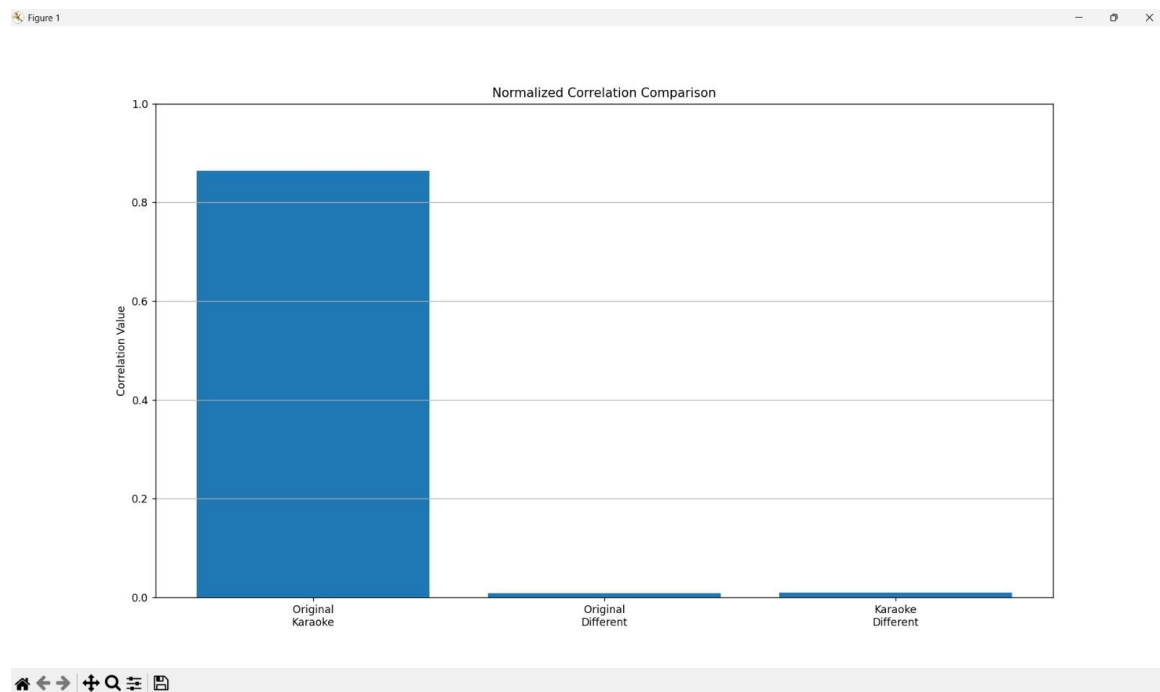
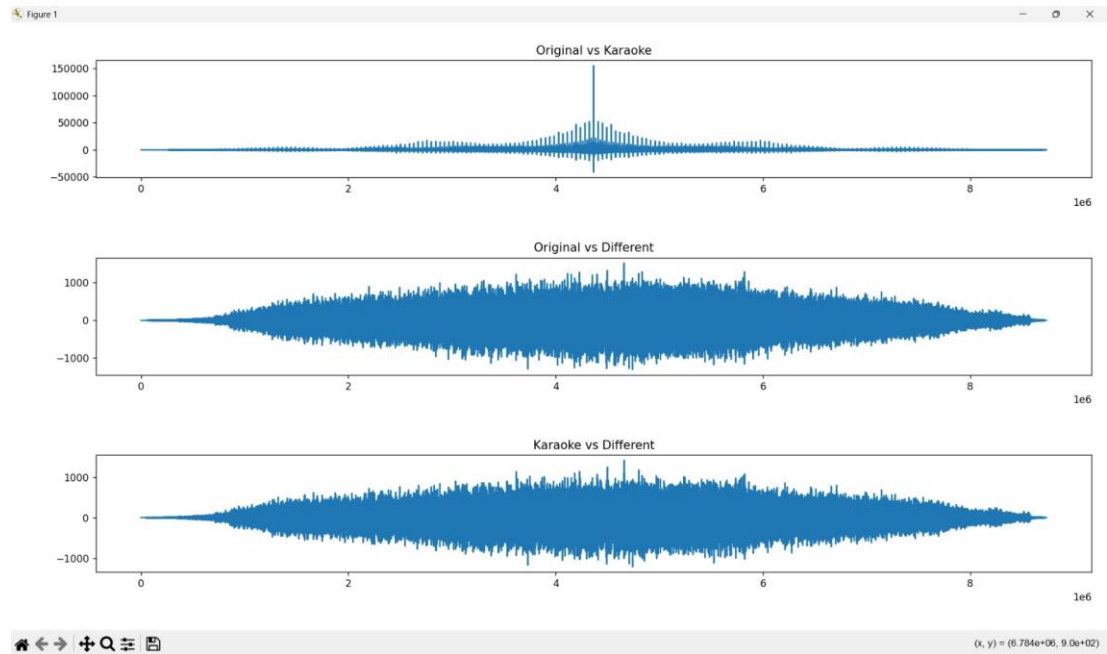
plt.grid(axis="y")

plt.show()
```

Output :



 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Digital Signal and Image Processing (01CT1513)	Aim : Compute and analyze the correlation between three audio tracks	
Open Ended Problem : 3	Date: 01/08/2026	Enrolment No: 92400133125



 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Digital Signal and Image Processing (01CT1513)	Aim : Compute and analyze the correlation between three audio tracks	
Open Ended Problem : 3	Date: 01/08/2026	Enrolment No: 92400133125

Code 2 :

```

import librosa

import numpy as np

import matplotlib.pyplot as plt

original, sr = librosa.load("Balam_Pichkari_Original.mp3", sr=None, mono=True)

karaoke, sr = librosa.load("Balam_Pichkari_karaoke.mp3", sr=None, mono=True)

different, sr = librosa.load("Aayi_Nai_-_Stree_2.mp3", sr=None, mono=True)

#it is take a 15 seconds to give response

duration = 15 # seconds

samples = sr * duration

original = original[:samples]

karaoke = karaoke[:samples]

different = different[:samples]

# for same length

min_len = min(len(original), len(karaoke), len(different))

original = original[:min_len]

karaoke = karaoke[:min_len]

different = different[:min_len]

corr_ok = np.corrcoef(original, karaoke)[0, 1]

corr_od = np.corrcoef(original, different)[0, 1]

corr_kd = np.corrcoef(karaoke, different)[0, 1]

print(f"Original vs Karaoke      : {corr_ok:.4f}")

```

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Digital Signal and Image Processing (01CT1513)	Aim : Compute and analyze the correlation between three audio tracks	
Open Ended Problem : 3	Date: 01/08/2026	Enrolment No: 92400133125

```

print(f"Original vs Different Song : {corr_od:.4f}")

print(f"Karaoke vs Different Song : {corr_kd:.4f}")

cross_ok = np.correlate(original, karaoke, mode='same')
cross_od = np.correlate(original, different, mode='same')
cross_kd = np.correlate(karaoke, different, mode='same')

plt.figure(figsize=(14, 8))

plt.subplot(3, 1, 1)

plt.plot(cross_ok)

plt.title("Cross Correlation: Original vs Karaoke")

plt.grid(True)

plt.subplot(3, 1, 2)

plt.plot(cross_od)

plt.title("Cross Correlation: Original vs Different Song")

plt.grid(True)

plt.subplot(3, 1, 3)

plt.plot(cross_kd)

plt.title("Cross Correlation: Karaoke vs Different Song")

plt.grid(True)

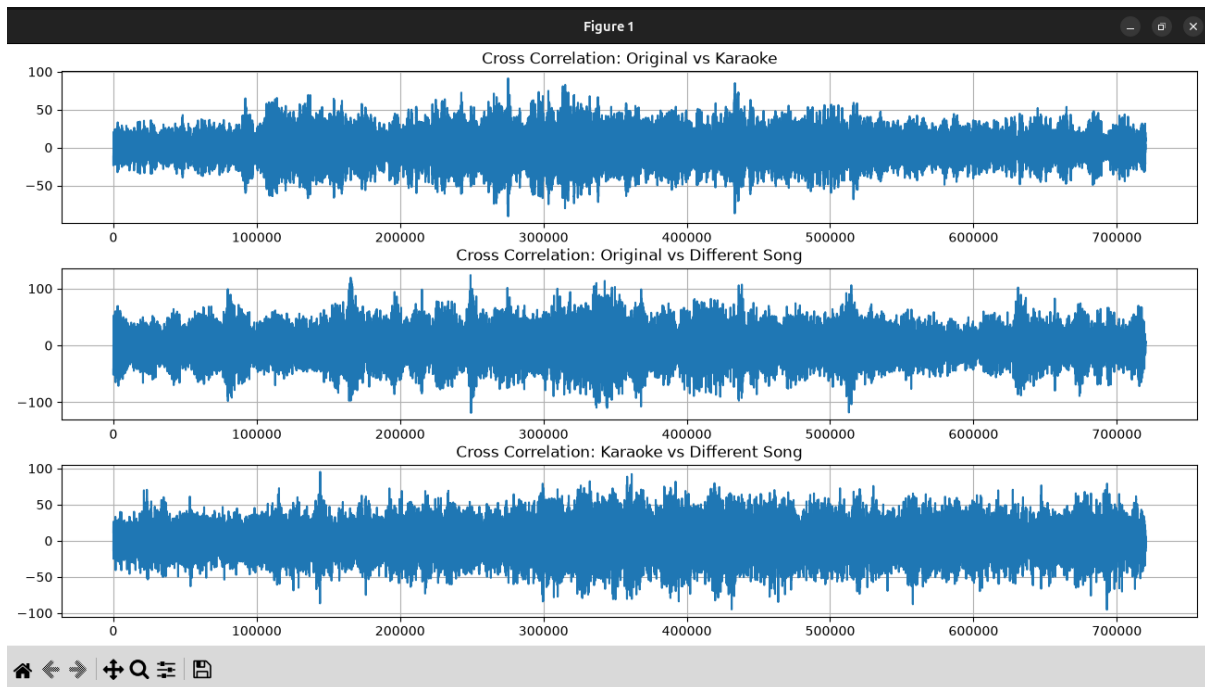
plt.tight_layout()

plt.show()

```

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Digital Signal and Image Processing (01CT1513)	Aim : Compute and analyze the correlation between three audio tracks	
Open Ended Problem : 3	Date: 01/08/2026	Enrolment No: 92400133125

Output :



Code 3 :

```
import librosa
import numpy as np
import matplotlib.pyplot as plt

# Load only first 5 seconds
original, sr = librosa.load(
    "Balam_Pichkari_Original.mp3",
```

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Digital Signal and Image Processing (01CT1513)	Aim : Compute and analyze the correlation between three audio tracks	
Open Ended Problem : 3	Date: 01/08/2026	Enrolment No: 92400133125

```

sr=None,

mono=True,

duration=5

)

karaoke, sr = librosa.load(

    "Balam_Pichkari_karaoke.mp3",

    sr=None,

    mono=True,

    duration=5

)

different, sr = librosa.load(

    "Aayi_Nai_-Stree_2.mp3",

    sr=None,

    mono=True,

    duration=5

)

min_len = min(len(original), len(karaoke), len(different))

original = original[:min_len]

karaoke = karaoke[:min_len]

different = different[:min_len]

print("Original vs Karaoke :", np.corrcoef(original, karaoke)[0,1])

```

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Digital Signal and Image Processing (01CT1513)	Aim : Compute and analyze the correlation between three audio tracks	
Open Ended Problem : 3	Date: 01/08/2026	Enrolment No: 92400133125

```

print("Original vs Different :", np.corrcoef(original, different)[0,1])

print("Karaoke vs Different :", np.corrcoef(karaoke, different)[0,1])

cross_ok = np.correlate(original, karaoke, mode="same")
cross_od = np.correlate(original, different, mode="same")
cross_kd = np.correlate(karaoke, different, mode="same")

plt.figure(figsize=(10,7))

plt.subplot(311)

plt.plot(cross_ok)

plt.title("Original vs Karaoke")

plt.grid()

plt.subplot(312)

plt.plot(cross_od)

plt.title("Original vs Different")

plt.grid()

plt.subplot(313)

plt.plot(cross_kd)

plt.title("Karaoke vs Different")

plt.grid()

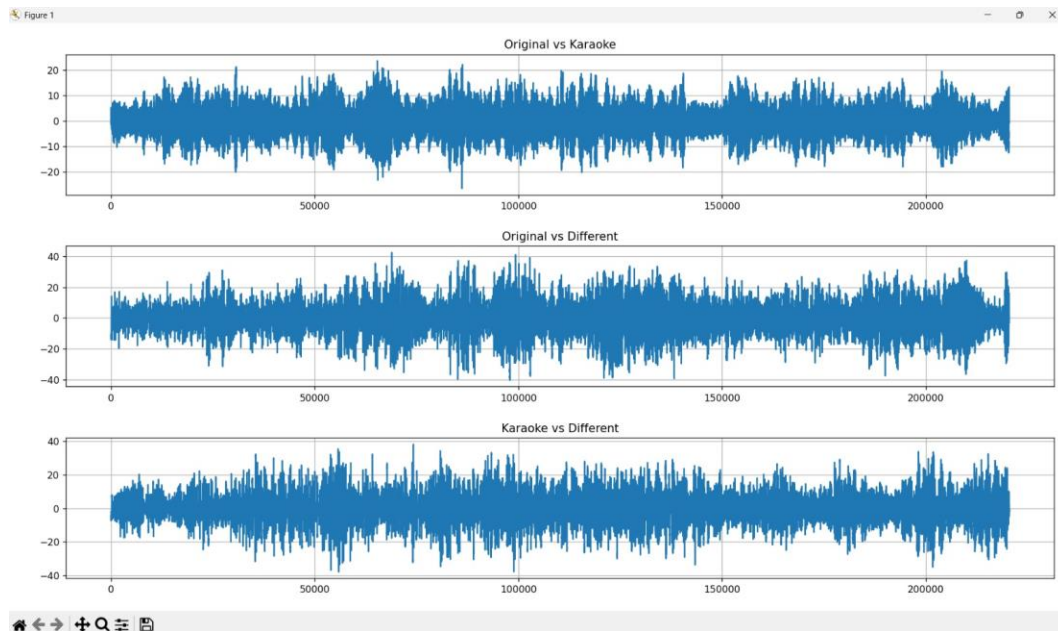
plt.tight_layout()

plt.show()

```

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Digital Signal and Image Processing (01CT1513)	Aim : Compute and analyze the correlation between three audio tracks	
Open Ended Problem : 3	Date: 01/08/2026	Enrolment No: 92400133125

Output :



Observation :

After performing the correlation analysis, the following observations were made:

- The Original Song and Karaoke Version produced a high cross-correlation because both contain almost the same instrumental music. Although the original version includes vocals, the background music is very similar in both tracks.
- The Original Song and the Different Song showed a much lower correlation. Since both songs have different melodies, instruments, rhythm, and vocals, their waveforms do not match well.

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Digital Signal and Image Processing (01CT1513)	Aim : Compute and analyze the correlation between three audio tracks	
Open Ended Problem : 3	Date: 01/08/2026	Enrolment No: 92400133125

Conclusion:

The correlation analysis showed that the original song and the karaoke version are the most similar, giving the highest correlation value because they share the same music. The different song had a much lower correlation with both tracks, indicating that it has a different melody and rhythm. This experiment shows that correlation is a simple and effective method for comparing the similarity between audio signals. The experiment successfully compared three different audio tracks using correlation techniques. The Original Song and Karaoke Version showed the highest similarity. The Original Song and Different Song showed very little similarity. The Auto-correlation of the original song produced the highest peak at zero lag.